THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
IKANA SECONDARY SCHOOL
FORM THREE MID-TERM EXAMINATION

033/1 **BIOLOGY 1**

Time: 3 Hours **YEAR: August 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section **A** carries **sixteen (16)** marks, section **B** carries **fifty-four (54)** marks and section **C** carries **thirty (30)** marks.
4. All writing should be in **blue** or **black** pen, except for diagrams that must be drawn in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **name** on every page of your answer sheet (s)

SECTION A (16 Marks)

Answer all questions in this section.

1. For each of the items (i – x), choose the correct answer from the alternatives and write its letter beside the item number in the answer booklet.

(i) Which respiratory surface is used for gaseous exchange in tadpoles?

A. Spiracles
B. Gills
C. Lungs
D. Skin
E. Book lung

(ii) Which of the following tools is used to magnify small objects?

A. Hand lens and Petri dish
B. Hand lens and watch glass
C. Microscope and watch glass
D. Microscope and hand lens
E. Measuring cylinder and beaker

(iii) What is the correct safety measure when giving First Aid to a wounded person?

A. Washing hands with soap
B. Wearing protective gloves
C. Calling an ambulance
D. Washing the wound
E. Drying the wound with a clean cloth

(iv) The following are features of prokaryotes EXCEPT:

A. Have nuclear material
B. Are very small
C. Have a nuclear membrane
D. Have a cell wall
E. Are single-celled organisms

(v) What are the end products of lipid digestion?

A. Glucose and fructose
B. Fatty acids and glycerol
C. Amino acids and fructose
D. Glucose and oxygen
E. Amino acids and glucose

(vi) What happens when someone moves from bright to dim light?

A. Pupil becomes large
B. Pupil becomes small
C. Circular muscles contract
D. Radial muscles relax
E. Radial muscles contract

(vii) In a food chain with grass, wildebeests, lions, and bacteria — what happens if lions are removed?

A. Bacteria remain the same
B. Wildebeest numbers decrease
C. Grass decreases
D. Grass increases
E. Wildebeest numbers increase

(viii) What happens if phloem tissue is destroyed in a green plant?

A. Absorption of water stops
B. Absorption of minerals stops
C. Transport of water and food stops
D. Transport of manufactured food stops
E. Oxygen transport stops

(ix) A mother fed her young child for three years on the following foods; meat, maize, Butter and milk. From which of the following deficiency diseases is the child likely to suffer?

A. Kwashiorkor
B. Marasmus
C. Rickets
D. Scurvy.
E. Obesity

(x) A scientific name consists of two words. To which groups do they belong?

A. Genus and class
B. Genus and order
C. Genus and species
D. Genus and family
E. Kingdom and phylum

2. Match the descriptions of the parts of heart in **List A** with their corresponding terminologies in **List B** by writing the letter of the correct response besides the item number in the answer sheet provided.

List A	List B
(i) A chamber of the heart with relatively thick walls.	A. Auricle
(ii) A valve that prevents blood backflow from right ventricle to right atrium.	B. Aorta
(iii) A valve that prevents blood backflow from pulmonary artery to right ventricle.	C. Bicuspid
(iv) Tissue that separates left and right chambers	D. Sinoatrial node
(v) A chamber with relatively thin walls	E. Myocardium
(vi) A small patch of muscle in the right atrium which sets the time and rate of cardiac muscle concentration	F. Semilunar
	G. Septum
	H. Tricuspid
	I. Ventricle

SECTION B (54 Marks)

Answer **all** questions in this section.

3. (a) In a car accident, many people were injured and felt pains all over their bodies. The victims were given First Aid before being taken to the hospital. What was the aim of administering First Aid to the accident victims? Give four points.

   (b) Puberty is the transition period from childhood to adulthood where the body undergoes different psychological and physiological changes. Give five physiological changes which occur in boys during puberty.

4. You have visited a school farm and observed that all maize plants have small yellowish leaves and show stunted growth.

   (a) Name four elements missing in the soil of the school farm.

   (b) What functions do the named elements play in crop production? Give one function for each element.

5. (a) Differentiate anaerobic from aerobic respiration under these features:

Feature	Aerobic respiration	Anaerobic respiration
(i) Site in the cell		
(ii) Substrates involved		
(iii)End products		
(iv) Amount of energy per glucose molecule.		

   (b) One of the features of living organisms is ability to exchange of oxygen and carbon dioxide across a respiratory surface. Explain two factors that affecting the rate of gaseous exchange in human being.

6. (a) Form three students were provided with the model of human skeleton as shown in the **figure 1 below**. Observe it carefully and help them to identify the parts labelled **A-F**.

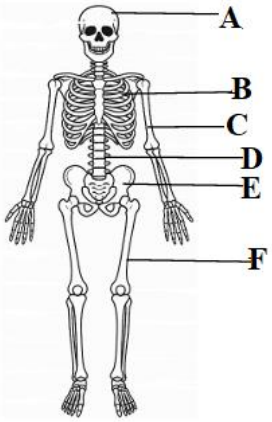


Figure 1

(b) How is the axial skeleton adapted to perform its function? Give three points

7. (a)Human beings are organisms whose body temperature is affected by environmental temperature. Briefly explain three ways that help human beings to survive in different weather conditions.
   (b) Explain the role of the two hormones produced by the pancreas gland in blood sugar regulation.
   (c) Explain how blood sugar can be regulated in the human body.

8. (a) Briefly explain the importance of the four excretory products of plants.
   (b) Draw a well diagram of the human urinary system and labelled the parts that perform the following functions:
      (i) Stores urine temporary before being discharged out of the body.
      (ii) Carries urine from the kidneys to the urinary bladder.
      (iii)Releases the urine to the outside of the body.
      (iv)Filters blood to remove urea, excess water and salts from the blood.

SECTION C (30 Marks)

Answer **only two (2)** questions in this section.

9. Drug abuse is among of the major problems in Tanzania. Justify this by discussing three causes and three ways of combating drug abuse in the country.

10. Explain four forms and two advantages of asexual reproduction.

11. As a Biology teacher, educate Form One students on the importance of studying Biology. Give six points.